

Semester Test 2

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Electricity and Electronics EBN 111

24 May 2023

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)			18

Lecturers: Prof X. Ye, Prof J.D. le Roux, Mr J. Thiselton**Assistant Lecturers:** S. Twala, G. Neeuwfan, B. Mahlakoana, M. Modiba**ECSA outcome assessment**

The following ECSA graduate attributes are assessed at entry level in this question paper: GA1 and GA2

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form, etc.
4. Read the instructions on the OCR form carefully.
5. Answers must include units where applicable.
6. Final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.

INSTRUCTIONS

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers to the OCR form.

- The Assessment ID is: **2023EBN111S02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Examples for complex numbers on OCR forms

5	∠	–	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	–	j	8						
j	1	5							Ω	
–	1	.	5	+	j	8	.	2	m	V

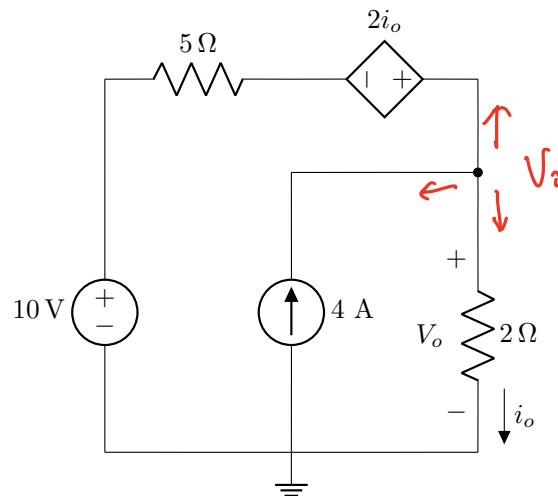
Rectangular:The *j* is at the start of the imaginary number.The *j* is in the CENTRE of the cell.The *j* does NOT cross or touch cell boundaries!**Polar:**

The degree sign ° is directly after the angle value.

Section 1

Questions 1 to 2 [4]

Use the principle of superposition to find the exclusive contributions from the independent sources to V_o .



01: The exclusive contribution from the independent voltage source, V_o^V [2]

02: The exclusive contribution from the independent current source, V_o^I [2]

01: switch off 4A $\uparrow$, it becomes an open circuit.

then: $-10 + 5i_o - 2i_o + 2i_o = 0$

$\Rightarrow i_o = 2A, V_o^V = 2 \times 2 = 4V.$

02: Switch off 10V $\oplus$, it becomes a short circuit

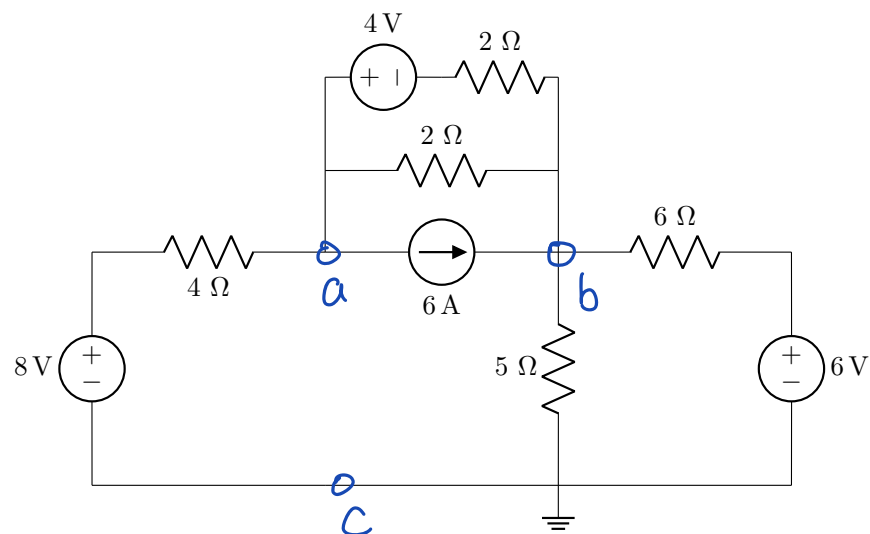
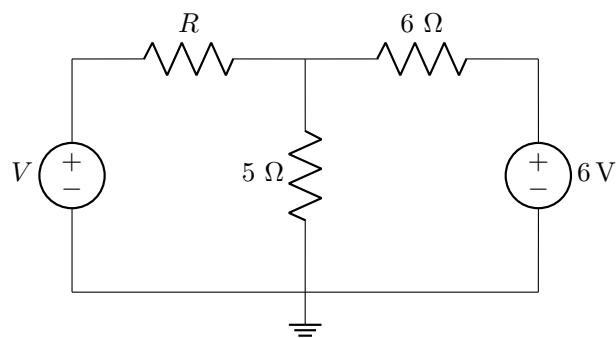
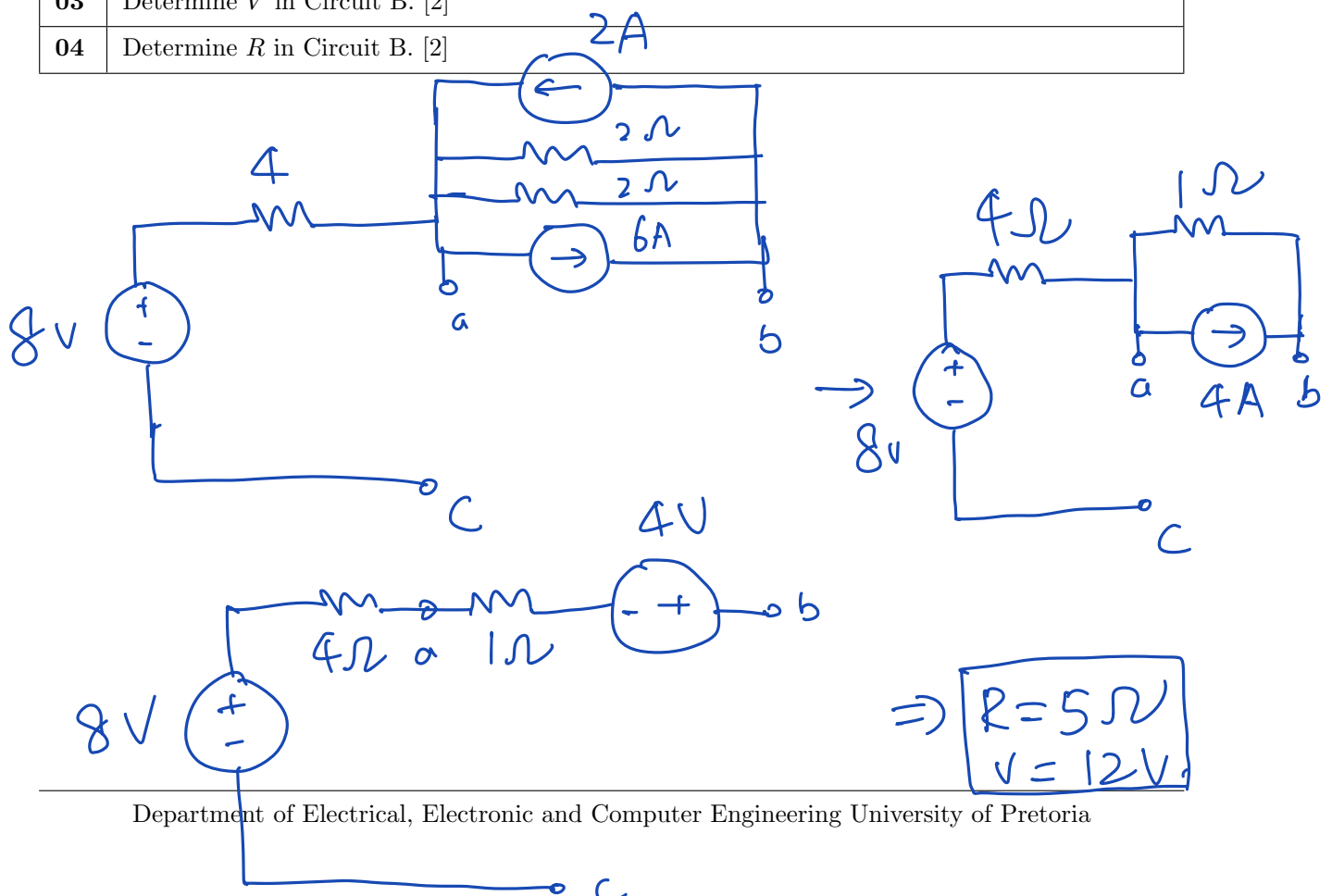
then at node V_o :

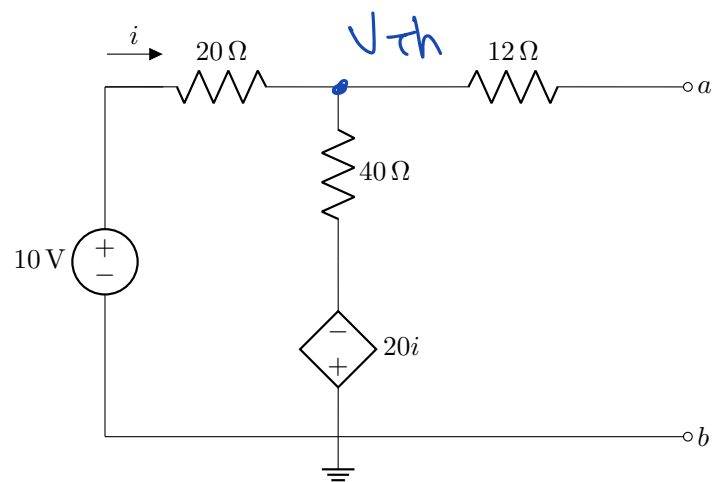
$$\frac{V_o - 2i_o}{5} - 4 + \frac{V_o}{2} = 0, V_o = 2i_o$$

$\Rightarrow V_o^A = 8V.$

Questions 3 to 4 [4]

Use source transformation to transform Circuit A to Circuit B.

Circuit A:**Circuit B:****03** Determine V in Circuit B. [2]**04** Determine R in Circuit B. [2]

Question 5 [2]Find the Thévenin equivalent voltage V_{Th} with respect to terminals a - b .05 | V_{Th} [2]

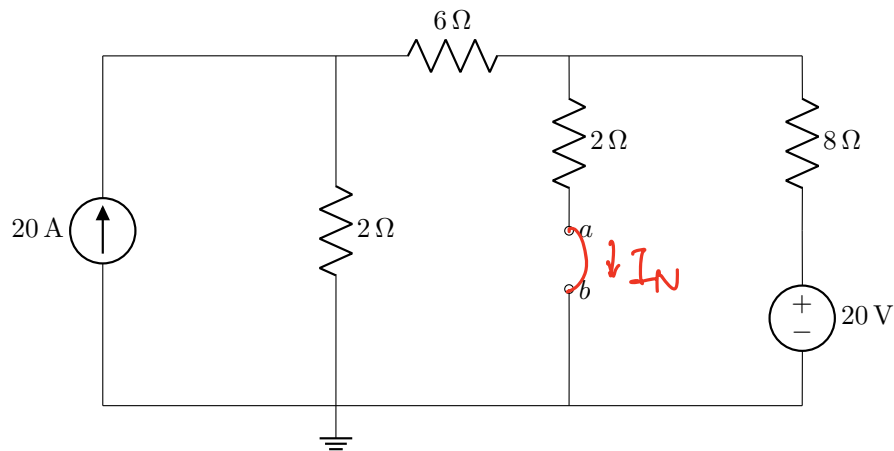
$$-10 + 20i + 40i - 20i = 0$$

$$\Rightarrow 40i = 10 \Rightarrow i = \frac{1}{4} \text{ A}$$

$$V_{Th} = 40i - 20i = 20i = 5 \text{ V.}$$

Questions 6 to 7 [4]

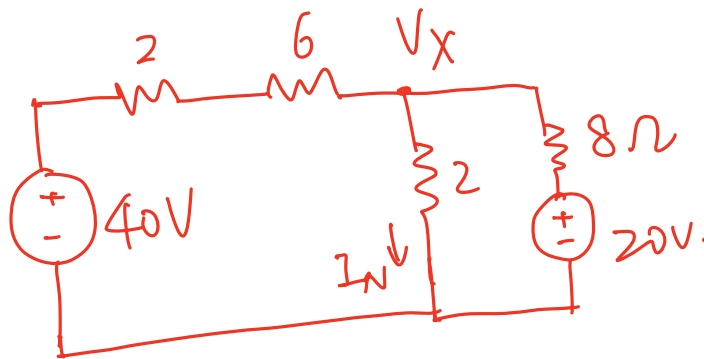
Determine the Norton equivalent current I_N , and the Norton equivalent resistance R_N with respect to terminals a - b .



06 I_N [2]

07 R_N [2]

$$R_N = 2 + (6 + 2) \parallel 8 = 6 \Omega$$



$$\frac{V-40}{8} + \frac{V}{2} + \frac{V-20}{8} = 0$$

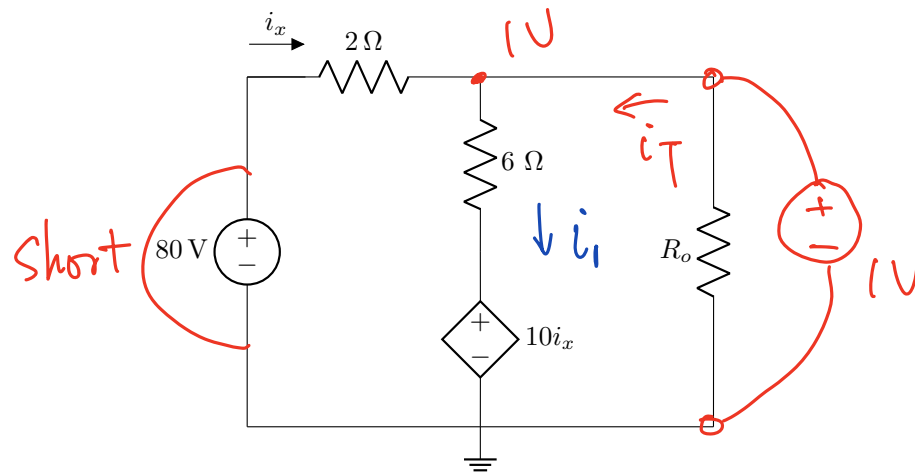
$$V-40 + 4V + V-20 = 0$$

$$\Rightarrow 6V = 60 \Rightarrow V = 10V$$

$$I_N = 5 A$$

Question 8 [2]

The resistor R_o in the circuit absorbs maximum power from the circuit. Calculate the value of R_o .



08 R_o [2]

$$i_x = -\frac{1}{2} = -0.5 \text{ A}$$

$$i_1 = \frac{1 - 10 \times (-0.5)}{6} = 1 \text{ A}$$

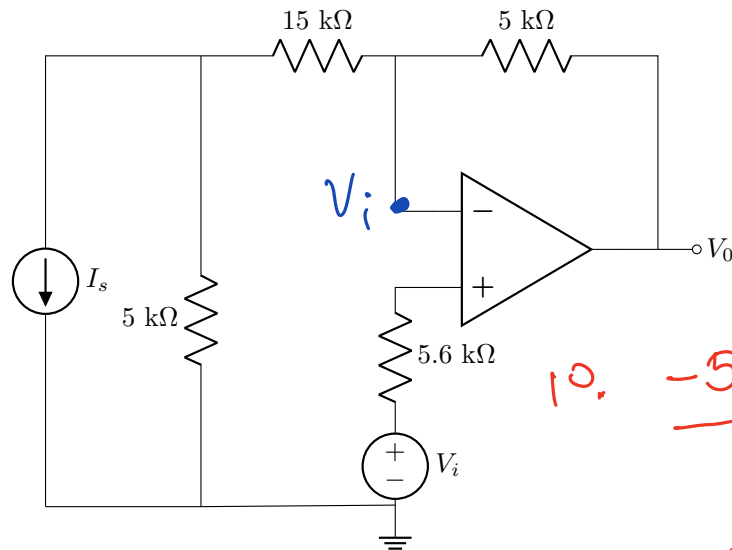
$$\Rightarrow i_T = i_1 - i_x = 1.5 \text{ A}$$

$$R_o = R_{Th} = \frac{1}{i_T} = \frac{1}{1.5} = \frac{2}{3} \Omega = 0.66667 \Omega$$

Total Section 1: [16]

Section 2

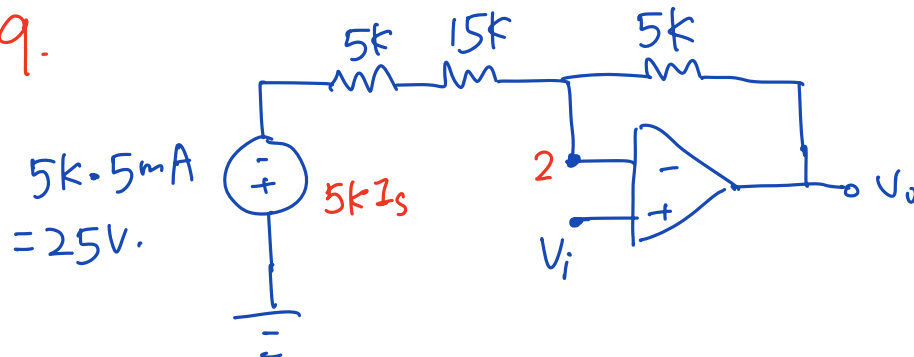
Questions 9 to 10 [4]



09 Given $I_s = 5 \text{ mA}$, and $V_i = 2 \text{ V}$, find V_o . [2]

10 If $V_o/V_i = 2$, find I_s when $V_i = 1 \text{ V}$. [2]

9.



$$\Rightarrow -5k I_s = -3$$

$$I_s = \frac{3}{5k}$$

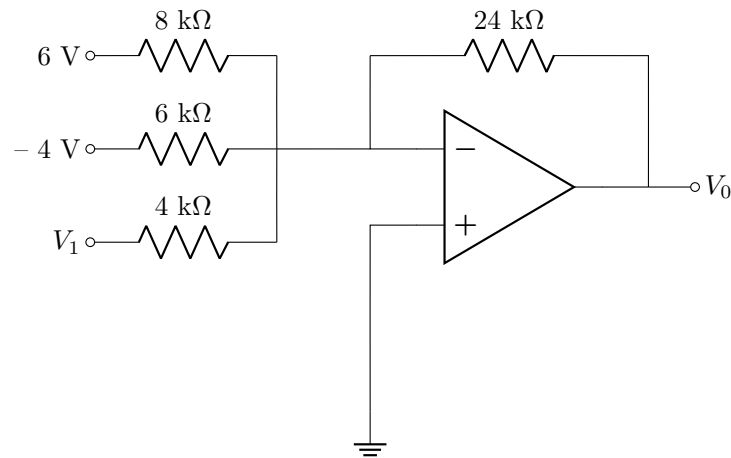
$$= 0.6 \text{ mA}$$

$$\frac{-25 - 2}{20k} = \frac{V_i - V_o}{5k} \Rightarrow -27 = 4V_i - 4V_o$$

$$\Rightarrow V_o = \frac{27 + 4V_i}{4}$$

$$= \frac{35}{4} = 8.75 \text{ V.}$$

Question 11 [2]



11 Find V₁ when V_o = 10 V. [2]

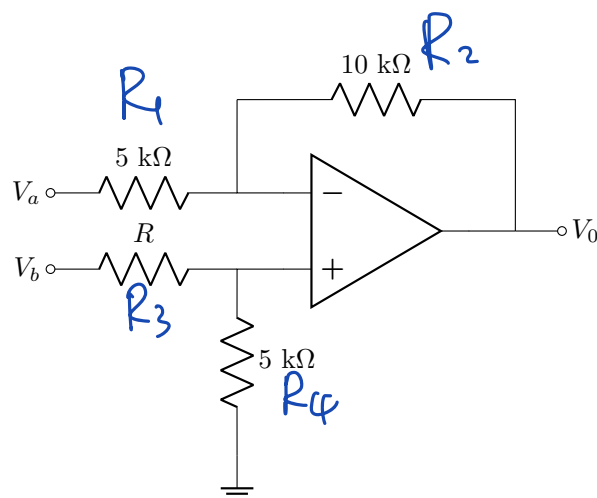
$$V_o = - \left(\frac{24k}{8k} \cdot 6 + \frac{24k}{6k} \cdot (-4) + \frac{24k}{4k} \cdot V_1 \right)$$

$$= 10 \text{ V}$$

$$- (3 \times 6 + 4 \times (-4) + 6V_1) = 10$$

$$-2 - 6V_1 = 10 \Rightarrow V_1 = -2 \text{ V.}$$

Question 12 [2]



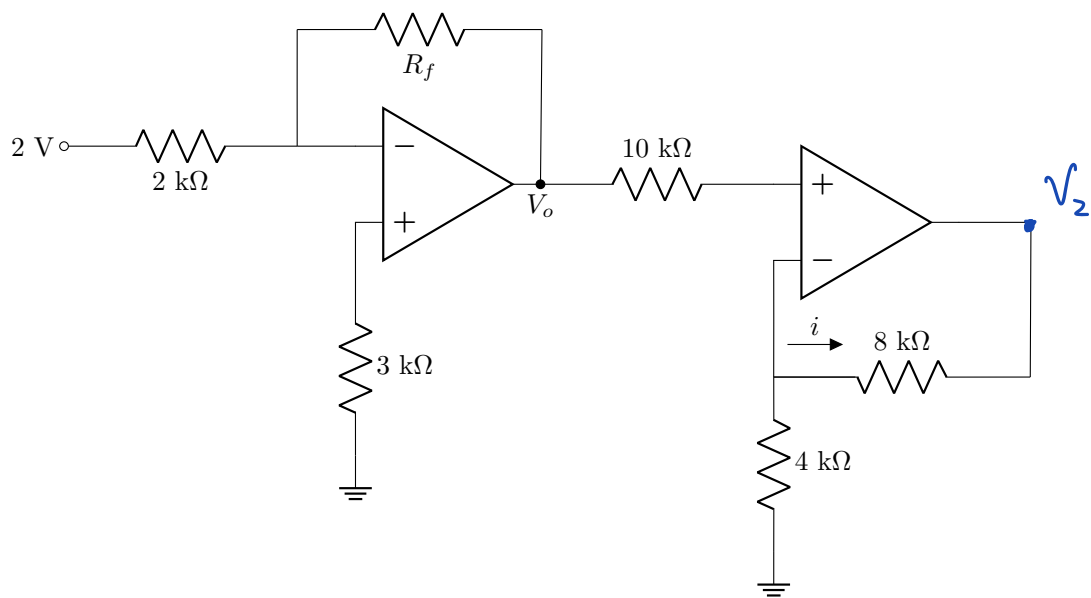
12 Determine R such that $V_o = 2(V_b - V_a)$. [2]

$$V_o = \frac{R_2}{R_1} \cdot \frac{\left(1 + \frac{R_1}{R_2}\right)}{\left(1 + \frac{R_3}{R_4}\right)} \cdot V_b - \frac{R_2}{R_1} V_a$$

$$= 2 \cdot \frac{\left(1 + \frac{1}{2}\right)}{\left(1 + \frac{R}{5}\right)} V_b - 2 V_a$$

$$\Rightarrow 1 + \frac{1}{2} = 1 + \frac{R}{5} \Rightarrow R = 2.5 \text{ k}\Omega.$$

Questions 13 to 14 [4]



13 If $R_f = 4 \text{ k}\Omega$, find V_o . [2]

14 If $V_o = 0.5 \text{ V}$, find i . [2]

$$13. V_o = -\frac{R_f}{2k} \cdot 2 = -4 \text{ V.}$$

$$14. V_2 = V_o \cdot \left(1 + \frac{8k}{4k}\right) = 1.5 \text{ V.}$$

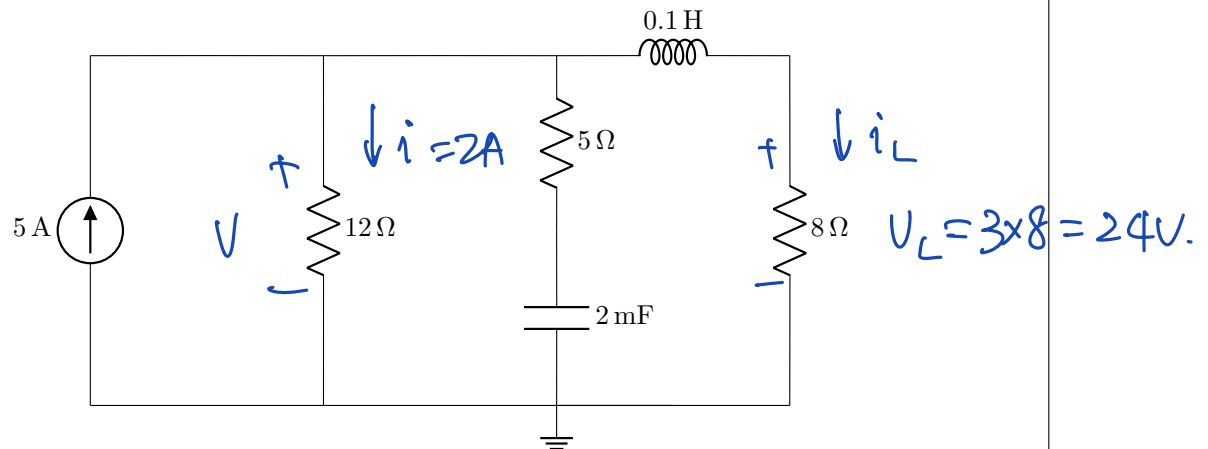
$$i = \frac{V_o - V_2}{8k} = \frac{0.5 - 1.5}{8k} = -\frac{1}{8} \text{ mA} = -125 \mu\text{A.}$$

Total Section 2: [12]

Section 3

Questions 15 to 16 [4]

The circuit below is under the steady state DC conditions.



15 Find the energy stored in the capacitor W_C . [2]

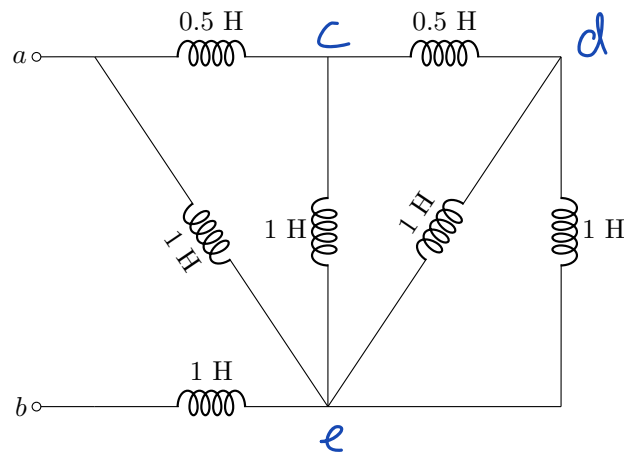
16 Find the energy stored in the inductor W_L . [2]

$$i_L = 5 \cdot \frac{12}{12+8} = 3A.$$

$$W_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 2 \times 10^{-3} \times 24^2 = 0.576J.$$

$$W_L = \frac{1}{2} L i^2 = \frac{1}{2} \times 0.1 \times 3^2 = 0.45J.$$

Question 17 [2]



- 17 Find the equivalent inductance L_{eq} with respect to the terminals $a-b$. [2]

$$L_{de} = 0.5 \text{ H}, \quad L_{ec} = (0.5 + 0.5) \parallel 1 = 0.5 \text{ H}$$

$$L_{ae} = 0.5 \text{ H}$$

$$L_{eq} = 1.5 \text{ H}$$

Question 18 [2]

The current, $i(t) = 100e^{-10t}$ mA, $t \geq 0$ flows through a 0.2 mF capacitor. The capacitor has an initial voltage $v(0) = 60$ V.

18 Find $v(t)$ when $t = 25$ ms. [2]

$$\begin{aligned} v(t) &= \frac{1}{C} \int_0^t i(t) dt + v(0) \\ &= \frac{1}{0.2 \times 10^{-3}} \int_0^t 100 e^{-10t} dt \times 10^{-3} + 60 \\ &= \frac{100}{0.2} \int_0^t e^{-10t} d\left(\frac{t}{10}\right) \cdot \frac{1}{-10} + 60 \\ &= -50 e^{-10t} \Big|_0^t + 60 \\ &= 50(1 - e^{-10t}) + 60 \\ v(25 \text{ ms}) &= 50(1 - e^{-10 \times 25 \times 10^{-3}}) + 60 \\ &= 71.060 \text{ V.} \end{aligned}$$

Question 19 [2]

Simplify the complex number **Z** in rectangular form.

$$\mathbf{Z} = \frac{30 + j40}{(2 - j1)\sqrt{6 - j8}}.$$

19 **Z** [2]

$$\sqrt{6 - j8} = \sqrt{10} \angle \frac{-53.13^\circ}{2} = 3.1623 \angle -26.565^\circ$$

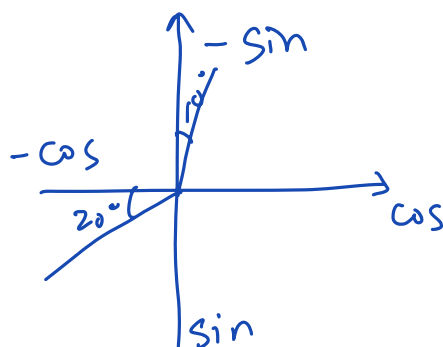
$$\bar{\mathbf{Z}} = 7.0710 \angle 106.26^\circ$$

Questions 20 to 21 [4]	
Given two sinusoids $v_1(t)$ and $v_2(t)$, where: $v_1(t) = -60 \cos(20t + 20^\circ),$ $v_2(t) = -120 \sin(20t - 10^\circ).$ The waveforms $v_1(t)$ and $v_2(t)$ can be represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.	
20	Choose the correct statement 1. v_1 leads v_2 by 120°. 2. v_2 leads v_1 by 120°. 3. v_1 leads v_2 by 30°. 4. v_2 leads v_1 by 80°. 5. None of the above is correct. Choose the correct statement number 1, 2, 3, 4, or 5 as your answer. [2]
21	Find $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$. [2]

$$V_1(t) = 60 \cos(20t - 160^\circ)$$

$$V_2(t) = 120 \cos(20t + 80^\circ)$$

Rotate V_1, V_2 clockwise by 30°
 then, $V_1' = 60 \angle -170^\circ$,
 $V_2 = 120 \angle 50^\circ$



Rotate V_1', V_2' clockwise by 180° ,
 then: $V_1'' = 60 \angle -10^\circ$,

$$V_2'' = 120 \angle -130^\circ$$

then V_1'' leads V_2'' by 120° .

Statement 1 is correct.

$$V = 60 \angle -160^\circ + 120 \angle 80^\circ$$

$$= 103.92 \angle 110^\circ.$$

Question 22 [2]

A 3Ω resistor is connected in series with a 0.1 H inductor. If a voltage signal $v(t) = 5\sin(40t + 0^\circ)\text{ V}$ is connected to the $R - L$ network in series, find the phasor current $\mathbf{I}_L$ through the inductor.

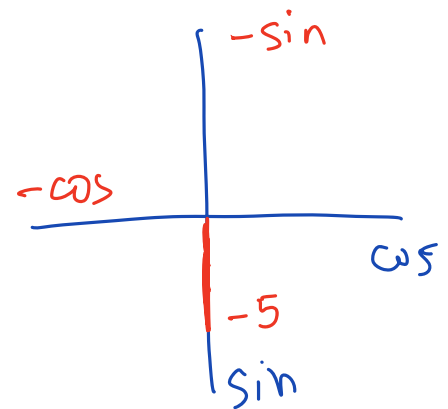
22 $\mathbf{I}_L$ [2]

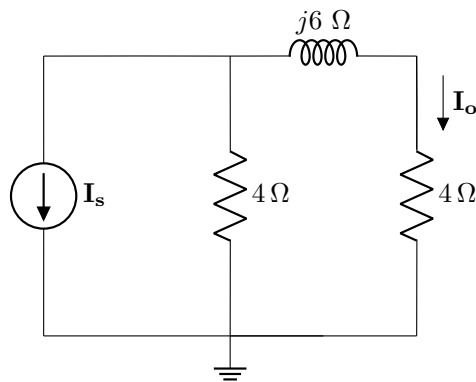
$$\omega = 40 \text{ rad/s.}$$

$$Z_L = j\omega L = j \times 40 \times 0.1 = j4\Omega.$$

$$\mathbf{I}_L = \frac{\bar{V}}{3 + j4} = \frac{5 \angle -90^\circ}{3 + j4}$$

$$= -0.8 - j0.6 \text{ A.}$$



Question 23 [2]**23** Find I_s if $I_o = 1\angle 0^\circ$ A. [2]

$$I_o = -I_s \cdot \frac{4}{8+j6} = 1$$

$$I_s = -\frac{8+j6}{4} = -2-j1.5 \text{ (A)}.$$

Total Section 3: [18]**Grand Total : [46]**

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	3	E	B	N	1	1	1	S	0	2
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Please round all irrational numbers to a minimum of 5 significant digits.

	Answer	Unit
01	$V_o^V = 4$	V
02	$V_o^I = 8$	V
03	$V = 12$	V
04	$R = 5$	Ω
05	$V_{Th} = 5$	V
06	$I_N = 5$	A
07	$R_N = 6$	Ω
08	$R_o = 0.66667$	Ω
09	$V_0 = 8.75$	V
10	$I_s = 0.6$	mA
11	$V_1 = -2$	V
12	$R = 2.5$	Ω
13	$V_o = -4$	V
14	$i = -125$	μA
15	$W_C = 0.576$	J
16	$W_L = 0.45$	J
17	$L_{eq} = 1.5$	H
18	$v(25 \text{ ms}) = 71.060$	V
19	$Z = 7.0710 \angle 106.26^\circ$	No unit required
20	correct statement: 1	No unit required
21	$V = 103.92 \angle 110^\circ$	No unit required
22	$I_L = -8 - j0.6$	A
23	$I_s = -2 - j1.5$	A